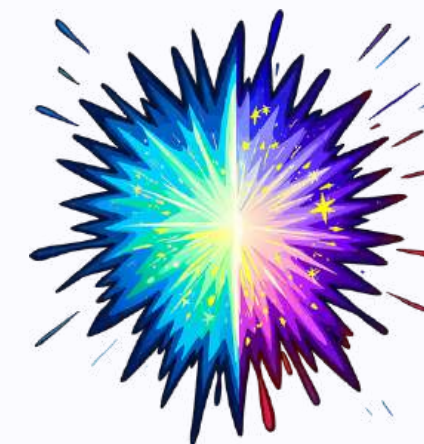


**SPECTROSCOPY —
FLAME TEST COLOURS
UNLOCKING NATURE'S
CHEMICAL RAINBOW**

**DISCOVER THE MAGIC OF
COLOURS IN FIRE!**





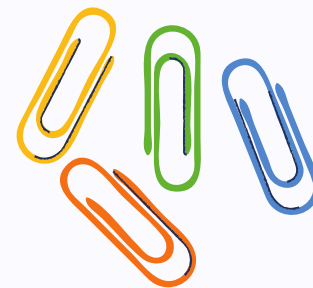
Diwali Fireworks Magic!

What Makes These Colours?



Have you ever wondered why firecrackers sparkle in so many different colours during Diwali?

Every burst of colour in the sky is a clue — a secret message from the elements! Let's find out what's hiding inside those sparkling lights!

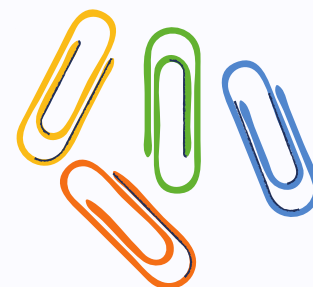




What is Spectroscopy?

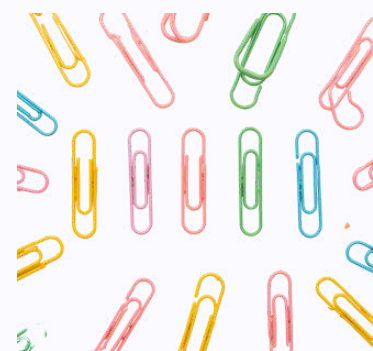
Spectroscopy means studying how light and matter dance together! When certain metals are heated, their tiny atoms get super excited and release energy as beautiful coloured light. Scientists use this to identify what elements are present — just by looking at the colour of the flame!

- ✓ Each metal has its own special "Jump & Flash Dance" — showing us a unique colour fingerprint!
- ✓ Heating metals = excited atoms = colourful light. That's spectroscopy in action!





Flame Colours!



Sodium

Bright Yellow flame!
Found in salt (NaCl).
Sodium's yellow glow is its special colour fingerprint!

Potassium

Lilac & Pale Violet flame!
Makes fireworks sparkle
with soft purple colours. A festive glow!

Copper

Blue-Green flame! Found
in copper pots & vessels.
An emerald spark with a beautiful glow!





Mystery Lab: Match the Colour!

Look at the flame colours and guess which element is which!

-  Bright Yellow → _____
-  Lilac / Pale Violet → _____
-  Blue-Green → _____
-  Brick Red → _____
-  Deep Crimson → _____

Element Bank: Sodium · Potassium · Copper ·
Calcium · Strontium



Sodium's Bright Yellow The Kitchen Star!

Sodium is found in common salt (NaCl) — the very same salt we use every day in our Indian kitchen! When sodium burns, it glows a brilliant, bright yellow colour.

Fun Fact: Next time you see a bright yellow flicker near the stove, that's sodium showing off its special colour fingerprint! Every element has its own unique glow.





Potassium's Lilac Flame The Festive Violet Glow

Potassium makes fireworks sparkle with soft, beautiful purple colours! When potassium burns, its atoms do a special "Jump & Flash Dance" and glow with a gentle lilac light.

 Fun Fact: Potassium's lilac flame adds magic to many Diwali crackers from Sivakasi! Each burst of violet sparkle in the night sky is potassium showing off its special colour fingerprint!

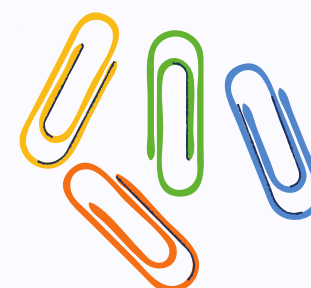




Copper's Blue-Green Flame!

Copper gives a beautiful blue-green flame — also called the Emerald Spark! When copper is heated, its atoms jump and flash in a stunning blue-green glow.

Fun Fact: Copper is used in many Indian cooking and water storage pots! The blue-green colour is copper's very own special fingerprint in the flame test.



Calcium's Brick Red Flame!

Calcium burns with a warm brick red flame — just like the glow of hot embers in a fire!
Each element has its own special colour fingerprint, and Calcium's is this beautiful deep red.

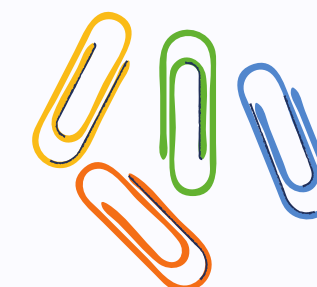
🦷 Fun Fact: Calcium is found in our bones, teeth, milk, and many foods we eat every day — and when heated, it glows brick red!

Look for this warm red glow in Diwali diyas and festive lamps!





Real-World Wonders!

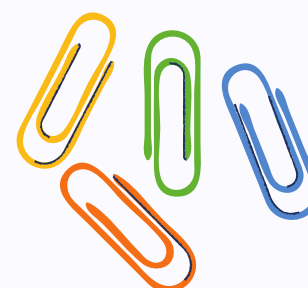




Quick Quiz!

Let's see how well you remember the flame colours! Look at the questions below and pick the right answer. Each element has its own special colour — can you match them all?

- ✓ Q1: Which element makes a bright yellow flame?
(a) Sodium (b) Copper (c) Calcium
- ✓ Q2: Potassium's flame colour is...
(a) Brick Red (b) Lilac (c) Blue-Green





THANK YOU! KEEP EXPLORING!

Every element shines with its own special light.

Ask your teacher for more science magic!

Spectroscopy — Flame Test Colours

